RLVR Follows a Frontier Set by the Base Model: Matched-Accuracy Evidence from Controlled Interventions and Public Checkpoints

Anonymous Authors[1]

Abstract

Comparisons of reinforcement learning with verifiable rewards (RLVR) at equal training steps mix learning speed with repeated-sampling coverage. We compare pass@16 at matched pass@1 with pre-registered decision rules and problem-bootstrap intervals. On Qwen2.5-Math-1.5B and 7B, the interpretable comparisons of more than twenty interventions—advantage rules, replay, larger groups, rollout allocation, human-solution guidance, process credit, update-geometry changes and teacher distillation at the tested dose—are consistent with progress along Dr. GRPO's pass@16-versus-pass@1 curve within each setting, not with a change of the curve. Training 6.7× longer raises pass@1 by 9–13 points without meeting the registered coverage criterion. A frozen per-problem law computed from the base model's success counts predicts this curve. Evaluated under one fixed protocol, 26 public checkpoints from other groups lie within +0.9 points of, or below, the law's prediction, including off-policy, external-supervision and random-reward recipes; the largest departures are downward and coincide with truncation. At matched pass@1, different methods show similar per-problem success profiles although their saved weight updates differ, and forcing a final answer recovers 16% of Dr. GRPO's pass@1 gain on the base model. Starting-distribution changes remain a candidate lever, and we register a prospective test of whether the law forecasts their effect.

[1]Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

1. Introduction

RLVR trains a language model on automatically checked answers (Shao et al., 2024; Guo et al., 2025). A large literature modifies how sampled answers are weighted (Liu et al., 2025b; Zhu et al., 2025; Walder & Karkhanis, 2025), which problems receive rollouts (Wang et al., 2026a), what is replayed (Zhan et al., 2025), or what external solutions are mixed in (Yan et al., 2025; Ma et al., 2025). Many report higher pass@k at a fixed number of steps. Yet RL-trained models are often beaten by their own base model at large k (Yue et al., 2025), and a method that progresses more slowly is less sharpened and therefore has higher pass@k at any fixed step.

We separate speed from coverage by comparing methods at matched pass@1: a method's pass@16 is compared with a reference curve interpolated at the same pass@1. All decision rules are registered before the corresponding runs, and intervals come from bootstrap draws over problems that recompute the reference in each draw. Our findings are:

1. One curve per setting. The interpretable registered comparisons of more than twenty interventions lie on Dr. GRPO's curve within ±2 points (Section 3). The pattern replicates at 1,024 tokens and at 7B with 2,048 tokens.
2. Longer and stronger training does not leave it (Section 4). 1,000-step runs reach pass@1 up to 70.7 while pass@256 changes by −1.0 to +0.8 points between steps 150 and 1,000.
3. Public checkpoints fall on or below the curve predicted from the base model (Section 5). Across 26 checkpoints and 38 evaluations, the largest residual against a frozen law is +0.9 points.
4. What is shared is behaviour, not a weight direction (Section 6). Methods solve similar problems at similar rates; their saved updates overlap only as much as two seeds of one method; part of the gain is learning to finish an answer.

5. Methods designed to move the curve did not (Section 7), including verified teacher solutions at the tested dose.

Section 9 discusses what can still move the frontier and the prospective tests now running.

2. Setup and protocol

For problem x with binary reward, p_x is the policy's success probability. Over N problems, $P_k = \frac{1}{N}\sum_x[1-(1-p_x)^k]$, estimated without bias from n_x samples with C_x successes as $\widehat{P}_k = \frac{1}{N}\sum_x[1-\binom{n_x-C_x}{k}/\binom{n_x}{k}]$ (Chen et al., 2021).

Proposition 2.1 (pass@k at matched pass@1 measures spread). $P_1 = \mathbb{E}_x p_x$ and $P_k = \mathbb{E}_x f_k(p_x)$ with $f_k(p) = 1-(1-p)^k$ strictly concave for $k > 1$; at fixed mean, a mean-preserving spread of $\{p_x\}$ lowers P_k.

Training. Dr. GRPO (Liu et al., 2025b): advantage $A_i = r_i - \bar{r}$ within a group, a sequence-sum loss with a constant normalizer, one update per batch of fresh rollouts (near on-policy), no KL term, AdamW at 10^{-6}, 16 problems $\times$ 8 rollouts per step, 150 steps, MATH training problems (Hendrycks et al., 2021). Qwen2.5-Math-1.5B (Yang et al., 2024) is trained at 512 or 1,024 response tokens, Qwen2.5-Math-7B at 2,048 (FSDP, four GPUs per run).

Evaluation. 511 held-out MATH test problems with audited answer keys, 32 samples at $T = 1$ (256 where stated), vLLM with per-problem sampling seeds and the training prompt; its agreement with the earlier Hugging Face evaluator was checked before use.

Matched-accuracy comparison. For a checkpoint with pass@1 a, the gain is $\widehat{P}_{16} - R(a)$, where R is the reference's piecewise-linear pass@16 at pass@1 a. The Dr. GRPO reference for a seed is the larger of its training trajectory (base and saved steps) and its step-150 temperature sweep ($T \in \{1, 1.15, 1.3\}$). Gains average eligible checkpoints within a seed, then seeds with equal weights; 2,000 problem-bootstrap draws recompute the reference; draws in which more than 10% of points are clamped make a contrast "not interpreted". Average equivalence requires the pooled and each seed's 90% interval within ± 2 points; moving the curve requires a pooled gain of at least $+2$, a pooled 95% interval above zero, positive estimates in every seed, and Holm correction where registered. Later registrations (20–22) also require each seed's 90% interval within ± 2 for equivalence; clamping is assessed within each seed. Intervals condition on the trained seeds.

A frozen per-problem law. Each problem's log-odds θ_x start at an empirical-Bayes estimate from the base model's 32-sample counts and follow $\dot{\theta}_x = p_x(1-p_x) + \tau\frac{1}{N}\sum_{x'} p_{x'}(1-p_{x'})$ with $\tau = -0.20$, fitted once on Dr. GRPO's 1.5B checkpoints and frozen. A checkpoint is placed where the law's mean success equals its observed pass@1; pass@16 is then a prediction. The law describes the curve conditional on pass@1; it does not predict training speed or endpoints.

Table 1. Registered matched-pass@1 differences in pass@16 (points; 95% intervals unless marked 90%). Reference: Dr. GRPO's trajectory-and-temperature envelope (B) unless stated. Validation pilots use one seed and 255 problems.

Comparison	Setting	Difference
NSR vs PSR	1.5B, 512, 3 seeds	−0.20 [−0.97, +0.55] (90%)
Coverage-weighted replay vs B	1.5B, 512, 3 seeds	+1.19 [+0.54, +1.79]
Allocation AC vs B	1.5B, 512, 3 seeds	+0.14 [−0.36, +0.58]
NSR vs PSR	1.5B, 1,024, 3 seeds	−0.25 [−0.72, +0.20]
PSR vs B (trajectory)	1.5B, 1,024, 3 seeds	+0.11 [−0.26, +0.49]
NSR vs B (trajectory)	1.5B, 1,024, 3 seeds	−0.15 [−0.62, +0.31]
Allocation AC vs B	1.5B, 1,024, 3 seeds	−0.12 [−0.60, +0.29]
Human-solution prefixes vs B	1.5B, 1,024, 3 seeds	−0.05 [−0.52, +0.37]
Self-generated prefixes vs B	1.5B, 1,024, 3 seeds	−0.11 [−0.55, +0.32]
Minimum-help credit vs B	1.5B, 1,024, pilot	−0.01 [−1.24, +0.89]
PSR vs B (trajectory)	7B, 2,048, 2 seeds	−0.09 [−0.48, +0.31] (90%)
NSR vs B (trajectory)	7B, 2,048, 2 seeds	−0.09 [−0.45, +0.27] (90%)
Allocation AC vs B	7B, 2,048, 2 seeds	−0.15 [−0.57, +0.16] (90%)

3. One curve within each setting

Table 1 collects the registered comparisons. Advantage rules (PSR, NSR, constant baselines, noise injection), rollout allocation by frontier value and its controls, human-solution and self-generated prefix guidance, and process credit all reach average equivalence or fall within ± 1.2 points with intervals including zero. Coverage-weighted replay gains $+1.19$, below the registered $+2$. The registered contrasts replicate at 1,024 tokens (all four contrasts) and at 7B with 2,048 tokens (three of four; the fourth fails the clamping guard). Figure 1 shows the 1,024-token setting.

The answer budget is part of the estimand. At 512 tokens, 48% of Dr. GRPO's answers are truncated; short-answer arms then appear above its curve. A supervised warm start on base-model successes gains $+4.77$ points [$+3.45$, $+6.07$] (90%) at 512 tokens but $+2.15$ [$+1.10$, $+3.20$] when the same checkpoints are evaluated at 1,024, meeting the registered criterion for "largely a budget effect"; a smaller positive gain remains. Human-solution warm starts at 512 tokens lie 5–6 points above the curve with 6–7% truncation; their registered contrast is not interpretable for lack of overlap.

4. Longer and stronger training

A natural objection is that 150 steps are too few to leave the base model's neighbourhood (Liu et al.,

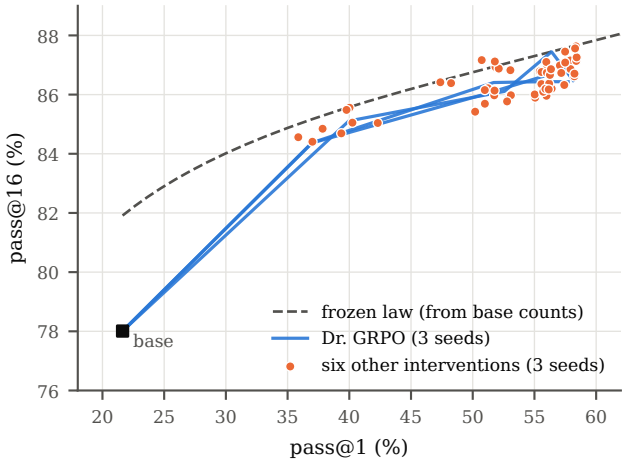


Figure 1. 1.5B at 1,024 tokens: Dr. GRPO's trajectories (three seeds) and checkpoints of six other interventions (PSR, NSR, allocation, human-solution and self-generated prefixes, three seeds each). Dashed: the frozen law computed from the base model's counts.

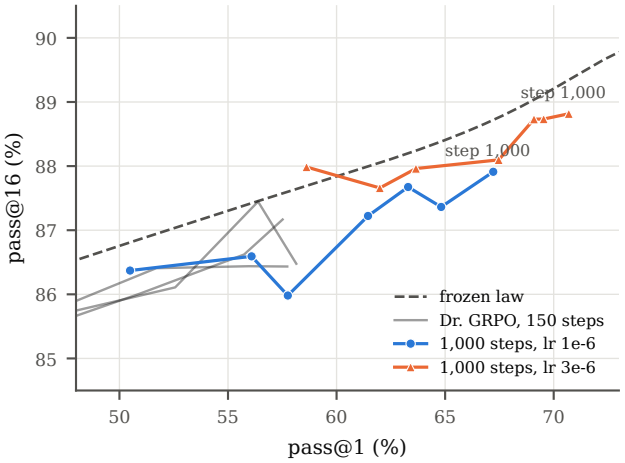


Figure 2. Dr. GRPO for 1,000 steps at two learning rates (checkpoints at steps 50–1,000) against the 150-step trajectories and the frozen law.

Table 2. Longer training (1.5B, 1,024 tokens). pass@256 from 256 samples; the base's pass@256 is 94.91. Law residual: pass@16 minus the frozen law at the checkpoint's pass@1.

	LH1 (10^{-6})	LH3 (3×10^{-6})
pass@1, step 150 → 1,000	57.8 → 67.2	63.7 → 70.7
pass@256 minus base, step 1,000	+1.96 [+0.39, +3.72]	+0.39 [−1.37, +2.15]
Holm-adjusted one-sided p	0.029	0.358
pass@256 change, step 150 → 1,000	+0.78	−0.98
Law residual at step 1,000	−0.81 [−2.67, +1.28]	−0.53 [−2.59, +1.76]
Spearman (base, arm), step 150 → 1,000	0.72 → 0.70	0.71 → 0.67

2025a). We trained Dr. GRPO for 1,000 steps (about four passes over the training problems) at learning rates 10^{-6} (LH1) and 3×10^{-6} (LH3), 1.5B at 1,024 tokens, and evaluated with 256 samples at steps 150, 500 and 1,000. The registered primary reading asks whether pass@256 at step 1,000 exceeds the base's by at least one point with a Holm-adjusted 95% interval above zero.

Neither arm meets the criterion (Table 2); LH1 is borderline (Holm-adjusted $p = 0.029$ against 0.025). Both arms move pass@1 well beyond Dr. GRPO's step-150 reference (+9.1 and +12.6 points; +9.4 and +7.0 over each arm's own step 150), so the registered verdict is "not supported" rather than inconclusive. The longer trajectories stay within 1.6 points of the frozen law (Figure 2), and the base model's difficulty order is largely preserved (Spearman 0.67–0.72). On the 26 problems with no success in 256 base samples, mean success rises from below 1% to 1.3–2.3%. Neither arm met the registered coverage criterion; LH1 nevertheless estimates a +1.96-point increase with borderline adjusted evidence. ProRL's prolonged training with KL control and resets reports a similar MATH500 pattern (pass@256 99.0 → 98.2) alongside gains on code; our test does not cover those additions.

5. Public checkpoints on the predicted curve

Our curve could be specific to our recipe. We evaluated published RLVR checkpoints trained from the same base models by other groups, under one protocol (our prompt and tokenizer, $T = 1$, 32 samples, pinned Hub revisions): 14 at 1.5B, evaluated at 1,024 and 3,072 tokens, and 14 at 7B at 2,048. They span Dr. GRPO and GRPO recipes (Liu et al., 2025b), SimpleRL, DAPO and RAFT++ variants, one-shot RLVR, rank-1 extrapolation (Wei et al., 2026), replay (Zhan et al., 2025), off-policy guidance from stronger-model traces (Yan et al., 2025), interleaved supervised fine-tuning (Ma et al., 2025), and correct- and random-reward reproductions (Shao et al., 2025). Of these 28 checkpoints (42 evaluations), two 1.5B checkpoints truncate more than 90% of answers at 1,024 tokens and fall outside the law's range at 3,072; the remaining 26 checkpoints give 38 law comparisons. The reference is the frozen law computed from the same base model's counts at the same budget; at 3,072 tokens it is the only reference available.

Figure 3 shows the result. Residuals (pass@16 minus the law) range from −7.6 to +0.3 points at 7B, −3.1 to +0.7 at 1.5B and 3,072 tokens, and −4.6 to +0.9 at 1.5B and 1,024 tokens. The largest observed residual is +0.89 (a point estimate). Of the 38 law comparisons, 11 are equivalent, 22 inconclusive and 5 depart downward; no interpretable evaluation meets the registered upward-departure criterion. The checkpoints furthest below it truncate heavily at our budget:

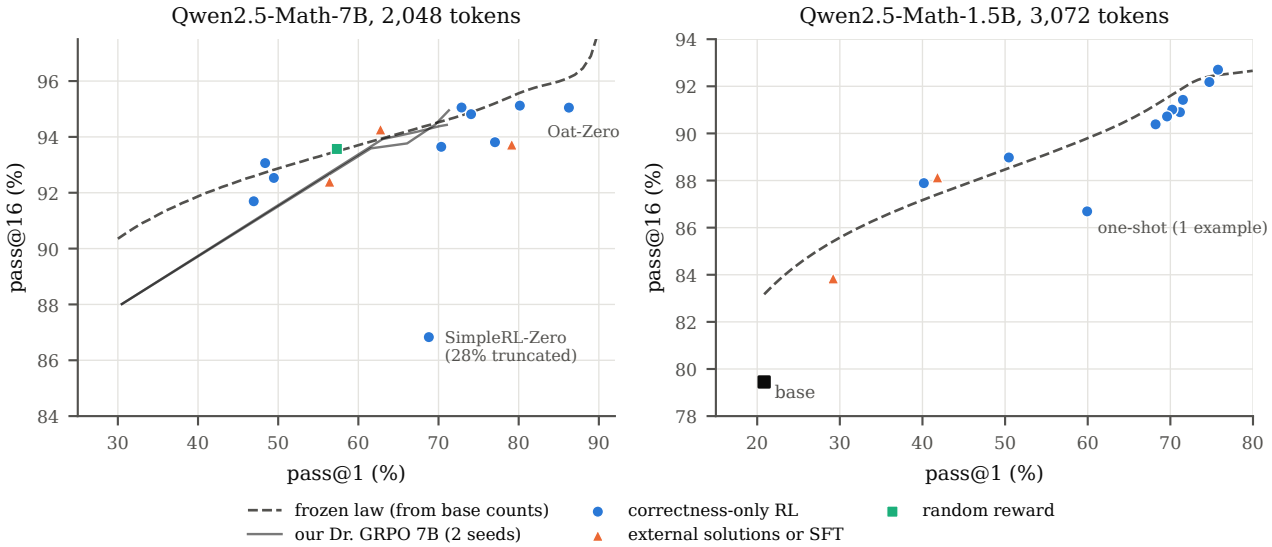


Figure 3. Public checkpoints against the frozen law computed from the base model at the same budget. Left: 7B at 2,048 tokens, with our Dr. GRPO trajectories. Right: 1.5B at 3,072 tokens. Colour and marker give the training-data provenance.

SimpleRL-Zero at 7B (28% truncated), and four 1.5B checkpoints at 1,024 tokens (30–52%); three of them (ExGRPO, LUFFY, ReLIFT) reach residuals of −1.57 to +0.72 at 3,072 tokens, while one-example RLVR remains at −3.10. Off-policy and external-supervision recipes are not exceptions: LUFFY at 7B −1.0, ExGRPO −0.3, ReLIFT −1.7, random reward +0.1, replay (RePO) −0.1 against +0.3 for its authors' own GRPO control. The registered per-checkpoint verdicts with Holm correction are reported in Appendix C. With 2,000 bootstrap draws and Holm correction, no checkpoint departs upward under either reference; the downward departures are the truncation-heavy checkpoints above and the one-example RLVR checkpoint at 1.5B. Against our empirical 7B trajectory the largest gain is one-shot RLVR's +1.8 [+0.95, +2.69] (90%, inconclusive); it lies where our 7B reference has no checkpoint between pass@1 30 and 62 and is linearly interpolated across the bend, and it is +0.3 against the law.

A fixed protocol is a limitation as well as a control: checkpoints trained with other prompts and longer budgets may perform better under their native settings. The test asks whether published RLVR recipes reach coverage beyond what the base model's counts predict under a common measurement; no interpretable evaluation met the registered upward-departure criterion.

6. What is shared

Per-problem profiles. At matched pass@1 (within 1.5 points), the root-mean-square per-problem difference in success rate between two methods is 6.6–6.7 points, the same as between two seeds of one method. A paired statistic—the excess squared per-problem distance of a method from Dr. GRPO over the distance between Dr. GRPO seeds—is +2.1% [−1.9%, +6.1%] of the seed-to-seed distance pooled over methods. Changes of the advantage rule leave a small signature (PSR +7.5% [+2.6%, +12.4%]); interventions that keep Dr. GRPO's advantages stay within seed variation. Mean success by base count is similar across methods at equal pass@1. This analysis is exploratory; per-problem outcomes still vary beyond what base success predicts (Yang et al., 2026), so the law is a conditional-mean description.

Saved updates fail the shared-direction criterion. Reported rank-1 structure of RLVR updates (Cai et al., 2025; Wei et al., 2026) suggested that methods move along one direction. On saved 1.5B checkpoints, per-matrix rank-1 approximations capture 4.7–5.6% of the update energy; the cosine between two methods' updates is 0.27–0.47, against 0.34 between Dr. GRPO seeds; PSR and NSR are the least similar pair (0.27). Removing each matrix's top singular component leaves 98% of the pass@1 gain. Top singular directions remain aligned (0.82–0.91), and expected-gradient alignment is unresolved. (Checkpoints are saved in bf16, which distorts the rank-1 component; the remainder is preserved with cosine 0.975.)

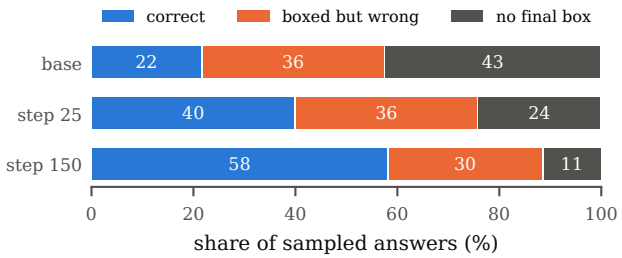


Figure 4. Answers of the 1.5B base and of Dr. GRPO (seed 8) at steps 25 and 150 at 1,024 tokens: correct, boxed but wrong, and without a final box (truncated or ended without one).

Finishing versus accuracy. At 1,024 tokens the base model leaves 42.5% of its answers without a final box (Figure 4). Between the base and step 25, correct answers rise by 18.4 points while unfinished answers fall by 18.3 and boxed-wrong answers are unchanged. By descriptive accounting, boxing more answers at the base's accuracy among boxed answers corresponds to +11.7 of the +36.2-point three-seed mean step-150 gain; accuracy among boxed answers also rises (37.6% → 65.3%). As an intervention, appending "Final answer: \boxed{" to every unfinished base sample and decoding 32 greedy tokens raises pass@1 from 22.0 to 27.7, against 22.3 for the same number of extra tokens without the cue: the cue-plus-extra-decoding intervention closes 16% of the gap to Dr. GRPO's step-150 pass@1, at slightly higher compute; it is not an estimate of RL's causal finishing contribution. Across problems, post-RL success is negatively associated with the base model's boxed-wrong rate (descriptive). Temperature is not a substitute: lowering the base model's temperature to 0.4–0.8 caps pass@1 near 29% with 45–53% truncation. Budget forcing at test time has prior art (Muennighoff et al., 2025); our contribution is measuring how much of RLVR's gain it accounts for.

7. Attempts to move the curve

Each of the following was designed from a diagnosis and pre-registered with a selection or decision rule.

More signal on rare problems. Dr. GRPO's update on a problem is zero unless its group mixes correct and incorrect answers. Allocation by frontier value raises the share of success-bearing groups among pilot-zero problems from 18–21% to 50–54%; at the two covered forecast coordinates (pass@1 37 and 40) the frozen law forecast +4.1 and the observed mean is −0.05 (the pooled registered estimate is +0.14). The tested allocation did not detect a registered frontier shift.

External solutions. Human-solution prefixes inside RL (−0.05), minimum-help process credit that credits failed prefixes by the smallest successful human hint (−0.01), do not move the curve; the warm-start advantage falls substantially with a larger evaluation budget but retains +2.15 points [+1.10, +3.20] (90%).

Update geometry and teacher signal (one-seed validation pilots; selection required +1.5): removing 75% of each update's component along the tracked dominant direction (+0.20 [−0.84, +1.19]); whitening problem weights by a sketched gradient Gram matrix (−0.04 [−1.21, +1.05]); adding the log-likelihood of verified 7B solutions on base-rare problems, 6.6 solutions per step (+0.31 [−1.05, +1.35]); and the same with the teacher gradient projected off the dominant direction (+0.22; −0.03 against plain teacher distillation). None qualifies. A gradient diagnostic with two independent rollout groups per problem found same-problem gradient cosines of 0.01; the registered whitening reading was inconclusive.

8. Related work

Yue et al. (2025) show that base models beat RLVR models at large k and that six RLVR algorithms perform similarly; we test that similarity at matched accuracy, across interventions and public checkpoints, and model it. The budgeted-search perspective (Sun et al., 2026) maps RL operating points to base search budgets with $N_{\text{base}} \approx \alpha N_{\text{RL}}^{\beta}$. In a restricted single-decoder adaptation fitted on our Dr. GRPO checkpoints, with a 256-sample base, it predicts held-out pass@16 with RMSE 1.98 points over 47 checkpoints, against 0.91 for Dr. GRPO's curve. DARS (Yang et al., 2025) reports matched-accuracy plots; we add registered intervals. Yang et al. (2026) show per-problem heterogeneity beyond base success, which our aggregate result tolerates. Rank-1 and subnetwork structure (Cai et al., 2025; Wei et al., 2026; Mukherjee et al., 2025) motivated our direction test. Reported departures include sharpness-aware updates (Kim et al., 2026), which reach equal rounded mean pass@1 with higher pass@256 on AIME but not on MATH500 (72.7 vs 72.5); BBG (Yuan et al., 2026); base anchoring (Zhou, 2026); ARMOR (Huang et al., 2026); and EAPO (Lu et al., 2026), mostly at fixed steps. Distillation and mid-training change what RL can reach (Kim et al., 2025; Wang et al., 2025; Zhang et al., 2025; Deshmukh et al., 2026), and pre-RL pass@large-k and held-out loss predict RL outcomes after SFT (Kang et al., 2025; Malladi et al., 2026). Reward noise as a change of clock has a theoretical predecessor (Rad et al., 2026); training on atomic skills can solve compositions the base never solved (Wang et al.,

2026b).

9. Discussion

Within the tested regimes, RLVR methods differ in how fast they move along a curve, and that curve is predicted by the base model's per-problem success counts. Three explanations were tested: lack of on-policy signal on unsolved problems (allocation and teacher solutions at the tested dose did not help), shared skills that lift problems in the base model's order (consistent with the per-problem and finishing analyses), and too-short training (not supported for plain Dr. GRPO over four passes). Starting-distribution changes remain a candidate lever; whether the frozen law forecasts their effect is unresolved, and it already failed after one supervised warm start (RMSE 3.56).

Running, registered tests. (i) Starting models fine-tuned on Qwen2.5-Math-72B-Instruct solutions for random, rarely solved and "confidently wrong" training problems at two doses, with the law's forecast of each start's post-RL curve frozen before RL, then RL from the starts with three seeds and a random-SFT control [P30]. (ii) Teacher distillation at 7B with the 72B teacher [P29]. (iii) Sharpness-aware RLVR on our curve [P31].

10. Limitations

Models are Qwen2.5-Math 1.5B and 7B (and OLMo-2 1B in earlier comparisons) on MATH-style problems with binary verification. Answer budgets of 512–3,072 tokens are part of the estimand; long chain-of-thought regimes, KL-regularized prolonged training, and other domains are not covered. Bootstrap intervals cover problem sampling conditional on the trained seeds. Average equivalence within ± 2 points permits smaller and checkpoint-specific differences. The law is a conditional-mean description fitted on Dr. GRPO; it fails after a supervised warm start (RMSE 3.56) and in base-zero strata at 7B ($+7$ to $+23$ points). Public checkpoints are evaluated under our protocol rather than their native one. The per-problem, weight-space and finishing analyses are exploratory.

References

Cai, Y. et al. On predictability of reinforcement learning dynamics for large language models. arXiv preprint arXiv:2510.00553, 2025.

Chen, M. et al. Evaluating large language models trained on code. arXiv:2107.03374, 2021.

Deshmukh, S. V. et al. Projected distillation: On learning to solve hard problems from close, correct targets. arXiv preprint arXiv:2610.27467, 2026.

Guo, D. et al. DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning. arXiv:2501.12948, 2025.

Hendrycks, D. et al. Measuring mathematical problem solving with the MATH dataset. arXiv:2103.03874, 2021.

Huang, K. et al. ARMOR: Stabilizing on-policy LLM RL with off-policy anchor samples. arXiv preprint arXiv:2607.10481, 2026.

Kang, F. et al. Quagmires in SFT-RL post-training: When high SFT scores mislead and what to use instead. arXiv preprint arXiv:2510.01624, 2025.

Kim, M. et al. Reinforcement learning vs. distillation: Understanding accuracy and capability in LLM reasoning. arXiv preprint arXiv:2505.14216, 2025.

Kim, Y. et al. Flat-RLVR: Sharpness-aware reinforcement learning with verifiable rewards for reasoning capacity in LLMs. arXiv preprint arXiv:2610.29281, 2026.

Liu, M. et al. ProRL: Prolonged reinforcement learning expands reasoning boundaries in large language models. In Advances in Neural Information Processing Systems, 2025a.

Liu, Z. et al. Understanding R1-Zero-like training: A critical perspective. arXiv:2503.20783, 2025b.

Lu, J. et al. Experience augmented policy optimization for LLM reasoning. arXiv preprint arXiv:2606.30420, 2026.

Ma, L. et al. Learning what reinforcement learning can't: Interleaved online fine-tuning for hardest questions. arXiv preprint arXiv:2506.07527, 2025.

Malladi, S. et al. TailSFT: Filtered fine-tuning improves post-training performance. arXiv preprint arXiv:2608.25756, 2026.

Muennighoff, N. et al. s1: Simple test-time scaling. arXiv preprint arXiv:2501.19393, 2025.

Mukherjee, S. et al. Reinforcement learning finetunes small subnetworks in large language models. arXiv preprint arXiv:2505.11711, 2025.

Rad, A. et al. Rate or fate? reinforcement learning with verifiable noisy rewards. arXiv preprint arXiv:2601.04411, 2026.

Shao, R. et al. Spurious rewards: Rethinking training signals in RLVR. arXiv:2506.10947, 2025.

Shao, Z. et al. DeepSeekMath: Pushing the limits of mathematical reasoning in open language models. arXiv:2402.03300, 2024.

Sun, W. et al. From base rollouts to RL reasoning: A budgeted search perspective. arXiv preprint arXiv:2609.01274, 2026.

Walder, C. and Karkhanis, D. Pass@k policy optimization: Solving harder reinforcement learning problems. arXiv:2505.15201, 2025.

Wang, T. et al. Where to spend rollouts: Hit-utility optimal rollout allocation for group-based RLVR, 2026a.

Wang, Z. et al. OctoThinker: Mid-training incentivizes reinforcement learning scaling. arXiv preprint arXiv:2506.20512, 2025.

Wang, Z. et al. New skills or sharper primitives? a probabilistic perspective on the emergence of reasoning in RLVR. arXiv preprint arXiv:2602.08281, 2026b.

Wei, Z. et al. You only need minimal RLVR training: Extrapolating LLMs via rank-1 trajectories. arXiv preprint arXiv:2605.21468, 2026.

Yan, J., Li, Y., Hu, Z., Wang, Z., Cui, G., Qu, X., Cheng, Y., and Zhang, Y. Learning to reason under off-policy guidance, 2025.

Yang, A. et al. Qwen2.5-Math technical report: Toward mathematical expert model via self-improvement. arXiv:2409.12122, 2024.

Yang, C. et al. RLVR is a kernel, not a function: Statistical inference for pass@k crossovers, 2026.

Yang, Z. et al. Depth-breadth synergy in RLVR: Unlocking LLM reasoning gains with adaptive exploration, 2025.

Yuan, S., Chen, J., Zheng, J., Li, M., Feng, L., Wang, D., Xiang, T., Liu, T., and An, B. Understanding diversity collapse in RLVR via the lens of overtraining, 2026.

Yue, Y. et al. Does reinforcement learning really incentivize reasoning capacity in LLMs beyond the base model? arXiv:2504.13837, 2025.

Zhan, R., Li, Y., Wang, Z., Qu, X., Liu, D., Shao, J., Wong, D. F., and Cheng, Y. ExGRPO: Learning to reason from experience, 2025.

Zhang, C. et al. On the interplay of pre-training, mid-training, and RL on reasoning language models. arXiv preprint arXiv:2512.07783, 2025.

Zhou, T. When RLVR shrinks the reasoning boundary: Diagnosing pass@k inversion. arXiv preprint arXiv:2607.20543, 2026.

Zhu, X., Xia, M., Wei, Z., Chen, W.-L., Chen, D., and Meng, Y. The surprising effectiveness of negative reinforcement in LLM reasoning. In Advances in Neural Information Processing Systems, 2025.

A. Expected updates

Theorem A.1 (Two moments of correctness-only rules). Let a rule draw answers i.i.d. from $\pi(\cdot \mid x)$, stop at a time $N \le n_{\max}$ that depends only on the rewards observed so far, and assign advantages that depend only on the rewards. Then $g = \sum_{i \le N} A_i \nabla \log \pi(y_i)$ satisfies $\mathbb{E}g = c(p)\nabla p$ and $\mathbb{E}\|g\|^2 = F_1 D_1(p) + F_0 D_0(p) + \|\nabla p\|^2 X(p)$, where F_r is the mean squared score norm of answers with reward r and c, D_1, D_0, X depend only on the rule and p.

Conditioning on the reward sequence fixes N and the advantages, and $\mathbb{E}[\nabla \log \pi \mid r = 1] = \nabla p / p$, $\mathbb{E}[\nabla \log \pi \mid r = 0] = -\nabla p / (1 - p)$. The expected update on each problem is collinear with ∇p_x, but the aggregate update through shared parameters need not be; the invariance we observe is empirical, not implied. Under Dr. GRPO a group without both a correct and an incorrect answer contributes nothing, with probability $p^n + (1 - p)^n$.

B. Registrations and verdicts

Reg.	Comparison	Verdict
9–11	NSR vs PSR on OLMo and Qwen; noise injection; temperature	Average equivalence; noise reproduces NSR's early lag; temperature sweep -1.42 $[-2.19, -0.68]$ (90%)
12, 14	Coverage-weighted and base-seeded replay	$+1.19$ and $+1.02$: below the $+2$ threshold
16 (check)	Answer budget	Warm-start gain $+4.77 \to +2.15$: largely a budget effect
17	Frozen per-problem law	RMSE criterion met on three test sets; fails after the warm start
18	Rollout allocation (AC, AV, HORA)	No rule met; AC forecast not confirmed
19	Human-solution warm starts	Rule 1 not interpretable (no overlap)
20	7B at 2,048 tokens	Three of four contrasts equivalent; one not interpreted
21	1.5B at 1,024 tokens	All four contrasts equivalent
22	Human vs self prefix guidance	Human prefixes on the curve; matched-problem contrast not interpreted
23	Minimum-help credit (pilot)	Not selected
26	Shared-direction diagnostics; OD, PW, TD, RTD pilots	Shared-direction criterion not met; no pilot selected
27	1,000-step training	Not supported for either learning rate
28	Public checkpoints; forced answer	No upward departure from the law (Appendix C)
29–31	72B teacher at 7B; start change with frozen forecasts; Flat-RLVR	[running]

C. Public checkpoints

Residual of pass@16 against the frozen law at the checkpoint's pass@1 (points; truncation in parentheses). Registered verdicts (2,000 draws, Holm within each size, budget and reference): against the law, 1.5B at 1,024 tokens 1 equivalent, 8 inconclusive, 3 departures down; 1.5B at 3,072 tokens 3, 8, 1; 7B 7, 6, 1; against Dr. GRPO's empirical curve (where covered), 1.5B 1, 1, 2 and 7B 5, 2, 1. No upward departure under either reference.

Checkpoint	1.5B, 1,024	1.5B, 3,072	7B, 2,048
Oat-Zero (Dr. GRPO paper)	−0.88 (11%)	+0.22 (1%)	−1.07 (1%)
Our Dr. GRPO, step 150	−1.6	–	−0.1
SimpleRL-Zoo / -Zero	–	–	−0.91 (3%) / −7.59 (28%)
RLHFlow GRPO / DAPO / RAFT++	−1.72 / −0.99	−0.83 / −0.24	– / – / −0.46
One-shot RLVR (1 example / 7.5k)	−3.83 (33%) / −1.01	−3.10 / −0.97	+0.33 / –
RLVR / RELEX (rank-1 extrapolation)	−0.12 / +0.64	−0.80 / −0.66	–
Zero-Solver; base GRPO	−1.49; +0.89	−0.52; +0.70	–
ExGRPO (replay)	−2.81 (30%)	+0.45	−0.29
LUFFY (stronger-model traces) / on-policy baseline	−4.63 (49%)	+0.72	−1.01 / −0.90
ReLIFT (interleaved SFT)	−3.28 (52%)	−1.57	−1.73
ExGRPO from LUFFY	–	–	+0.34
Correct / random reward (rethink RLVR)	–	–	−1.39 / +0.08
RePO / its GRPO control (DeepMath)	protocol failure	outside the law's range	−0.07 / +0.28